

HOMEWORK 2

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1. Let $\varphi : G \rightarrow H$ be a group homomorphism. Define the kernel of φ to be

$$\ker \varphi = \{g \in G : \varphi(g) = e_H\}.$$

Prove that $\ker \varphi$ is a subgroup of G and that $\text{im } \varphi$ (i.e. the range of φ) is a subgroup of H .

Proof. Let $g, h \in \ker \varphi$. Then $\ker \varphi$ is nonempty since $\varphi(e_G) = e_H$, and

$$\varphi(gh^{-1}) = \varphi(g)\varphi(h^{-1}) = \varphi(g)\varphi(h)^{-1} = e_H \cdot e_H^{-1} = e_H,$$

so $\ker \varphi$ is a subgroup of G .

Now let $g, h \in \text{im } \varphi$. Then $g = \varphi(x)$ and $h = \varphi(y)$ for some $x, y \in G$. Thus

$$gh^{-1} = \varphi(x)\varphi(y)^{-1} = \varphi(x)\varphi(y^{-1}) = \varphi(xy^{-1}),$$

so $gh^{-1} \in \text{im } \varphi$. Since $e_H = \varphi(e_G)$, $\text{im } \varphi$ is nonempty, so $\text{im } \varphi$ is a subgroup of H . □

2. Let $\varphi : G \rightarrow H$ be a group homomorphism, fix $g \in G$ and set $h = \varphi(g)$. Prove that the fiber of φ over h satisfies

$$\varphi^{-1}(h) = \{gk : k \in \ker \varphi\} = \{kg : k \in \ker \varphi\}.$$

Proof. To show $\varphi^{-1}(h) \subset \{gk : k \in \ker \varphi\}$, let $x \in \varphi^{-1}(h)$. Then

$$\varphi(g^{-1}x) = \varphi(g)^{-1}\varphi(x) = h^{-1}h = e_H,$$

so $g^{-1}x \in \ker \varphi$, and $x = g(g^{-1}x)$.

To show $\varphi^{-1}(h) \supset \{gk : k \in \ker \varphi\}$, let $k \in \ker \varphi$ and observe that

$$\varphi(gk) = \varphi(g)\varphi(k) = h \cdot e_H = h.$$

The proof that $\varphi^{-1}(h) = \{kg : k \in \ker \varphi\}$ is entirely analagous. □

3. Prove that each of the following subsets of $\text{GL}_n(F)$ (where F is a general field) are subgroups.

(a) $B_n(F) = \{M \in \text{GL}_n(F) : M \text{ is upper triangular}\}$

(b) $\text{SL}_n(F) = \{M \in \text{GL}_n(F) : \det(M) = 1\}$

- (c) $O_n(F) = \{M \in \text{GL}_n(F) : M \text{ is orthogonal}\}$
- (d) $H_n(F) = \{M \in \text{GL}_n(F) : M \text{ is upper triangular and has 1s along the diagonal}\}$

Proof. **UNFINISHED**

- (a) Let M_1, M_2 be upper triangular matrices. The inverse of an upper triangular matrix is itself upper triangular, so $M_1 M_2^{-1}$ is the product of two upper triangular matrices, which is known to be upper triangular.

□

4.

- (a) Let $\varphi : G \rightarrow H$ be a group homomorphism. Prove that for all $x \in G$ of finite order, the order of $\varphi(x)$ divides the order of x .
- (b) Use the previous problem to show that the order of an element does not change under an isomorphism.
- (c) Conclude that the only group homomorphism from $(\mathbb{Z}/n\mathbb{Z}, +)$ to $(\mathbb{Z}, +)$ is the trivial homomorphism (i.e. $\varphi(\bar{k}) = 0$ for all $\bar{k} \in \mathbb{Z}/n\mathbb{Z}$).

Proof.

- (a) Let $|x| = n$ and $|\varphi(x)| = m$. Certainly, $m \leq n$ since $\varphi(x)^n = \varphi(x^n) = \varphi(e_G) = e_H$. Then we can write $n = m\ell + r$, where $\ell, r \in \mathbb{Z}^{\geq 0}$ and $0 \leq r < m$. Thus

$$e_H = \varphi(e_G) = \varphi(x^n) = \varphi(x^{m\ell+r}) = \varphi(x)^{m\ell} \varphi(x)^r = (\varphi(x)^m)^\ell \varphi(x)^r = e_H \varphi(x)^r = \varphi(x)^r.$$

Since $r < n = |x|$, $r = 0$, so $m \mid n$.

- (b) If $\varphi : G \rightarrow H$ is an isomorphism, then given $g \in G$, $|\varphi(g)|$ divides $|g|$, and $|g| = |\varphi^{-1}(\varphi(g))|$ divides $|\varphi(g)|$ since φ^{-1} is a homomorphism. Thus $|g| = |\varphi(g)|$.
- (c) In $\mathbb{Z}/n\mathbb{Z}$, every element has finite order, and the only element of finite order in $\mathbb{Z}$ is 0. Part (a) implies that elements of finite order have an image of finite order under homomorphism, so if $\varphi : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}$ is a homomorphism, $\varphi(x) = 0$ for all $x \in \mathbb{Z}/n\mathbb{Z}$.

□

5. Let G and H be abelian groups, and let $\text{Hom}(G, H)$ be the set of all group homomorphisms from G to H . Prove that this set forms an abelian group under the operation $(\varphi + \psi)(x) = \varphi(x) + \psi(x)$.

Proof. We know $\text{Hom}(G, H)$ is nonempty since it contains the trivial homomorphism $t : x \mapsto e_H$. In fact, t is the identity since given $f, g, h \in \text{Hom}(G, H)$ and $x \in G$

$$(f + t)(x) = f(x) + t(x) = f(x) + e_H = f(x) = e_H + f(x) = t(x) + f(x) = (t + f)(x).$$

Also, $-f : x \mapsto -f(x)$ is the inverse of f since

$$(f + -f)(x) = f(x) + (-f(x)) = e_H = (-f(x)) + f(x) = (-f + f)(x).$$

Given $f, g, h \in \text{Hom}(G, H)$ and $x \in G$, we have associativity as well:

$$((f+g)+h)(x) = (f+g)(x) + h(x) = f(x) + g(x) + h(x) = f(x) + (g+h)(x) = (f+(g+h))(x).$$

Finally, we have commutativity by

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x).$$

Note that this actually only depends on H being abelian. □

6. Let H be a subgroup of a group G , and let $x \in G$. Prove that

$$xHx^{-1} = \{xhx^{-1} : h \in H\}$$

is a subgroup of G . Also, prove that G acts on the set of all subgroups of G by conjugation.

Proof. Since $e = xx^{-1} = xex^{-1}$, $e \in xHx^{-1}$, so xHx^{-1} is nonempty. Then let $g, h \in H$, and see that

$$xgx^{-1}(xhx^{-1})^{-1} = xgx^{-1}xh^{-1}x^{-1} = xgh^{-1}x^{-1},$$

and $gh^{-1} \in H$ since H is a subgroup, so the subgroup criterion is satisfied.

We proceed to show that G acts on the set of subgroups of G by conjugation. The first paragraph shows that conjugation does indeed send subgroups to subgroups, so it remains to check the behavior of identities and compositions of conjugations. Let H be a subgroup of G . Then, $eHe^{-1} = H$, and for $g, h \in G$,

$$g(hHh^{-1})g^{-1} = (gh)H(h^{-1}g^{-1}),$$

so conjugation does indeed act on the set of subgroups. □

7. Let G act on a set X , and let $x \in X$. Prove that the stabilizer of x :

$$\text{Stab}_G(x) = \{g \in G : g \cdot x = x\}$$

is a subgroup of G . Then prove that for any $g \in G$ and $x \in X$, one has that

$$\text{Stab}_G(g \cdot x) = g\text{Stab}_G(x)g^{-1}$$

where the right hand side is defined by Problem 6 above.

Proof. We know that $e \in \text{Stab}_G(x)$ because $e \cdot x = x$ by definition of an action. Given $g, h \in \text{Stab}_G(x)$, we have

$$g^{-1} \cdot x = g^{-1} \cdot (g \cdot x) = (g^{-1}g) \cdot x = e \cdot x = x,$$

and

$$(gh) \cdot x = g \cdot (h \cdot x) = g \cdot x = x,$$

so $\text{Stab}_G(x)$ is closed under multiplication and taking inverses, and thus is a subgroup of G . □

8. Let $GL_n(\mathbb{R})$, $SL_n(\mathbb{R})$, and $O_n(\mathbb{R})$ be the general linear group, the special linear group, and the orthogonal group, respectively (see their definitions from Problem 3). Each of these groups acts on $\mathbb{R}^n$ via matrix multiplication. Determine the orbits of the action in each case (and prove your answer is correct).

Proof. **UNFINISHED** For $GL_n(\mathbb{R})$, the orbits are $\mathbb{R}^n \setminus \{0\}$ and $\{0\}$. Indeed, no nonzero vector in $\mathbb{R}^n$ can be taken by an invertible matrix M to 0 since then 0 would be an eigenvalue of M , making the determinant 0. On the other hand, so long as at least one component x_i of a given $x \in \mathbb{R}^n$ is nonzero, then it can be transformed into any other nonzero vector $y \in \mathbb{R}^n$ by the matrix whose i^{th} column is y/x_i (as a column vector) and whose other entries are all zero.

For $SL_n(\mathbb{R})$, □

9. Let G and H be groups, and let $\mathbf{G}$ and $\mathbf{H}$ be the categories with one object whose composition law is the group law as considered in the previous homework set. Show that the data of a group homomorphism $\varphi : G \rightarrow H$ contains the same information as a functor $F : \mathbf{G} \rightarrow \mathbf{H}$.

Proof. A functor $F : \mathbf{G} \rightarrow \mathbf{H}$ sends the single object of $\mathbf{G}$ to the single object of $\mathbf{H}$, and it sends morphisms in $\mathbf{G}$ to morphisms in $\mathbf{H}$. Thus, since the elements of G and H are identified via isomorphism with the morphisms in $\mathbf{G}$ and $\mathbf{H}$ respectively, we want to check that the preservation of the group operation (i.e. the data of a group homomorphism $G \rightarrow H$) maps on directly to the composition of corresponding morphisms by the functor F (i.e. the morphism data of F).

Indeed, this falls right out of the definition. To see this, take two morphisms $f, g \in G$, and we get $F(g \circ f) = F(g) \circ F(f)$ and $F(\text{id}_{*_{\mathbf{G}}}) = \text{id}_{*_{\mathbf{H}}}$ by definition. Now identifying f and g with their corresponding elements of G by a slight abuse of notation, we see that $\varphi(fg) = \varphi(f)\varphi(g)$ is the same statement as above for any homomorphism $\varphi : G \rightarrow H$, and the identity in G gets mapped to the identity in H as a consequence. □